

MACHINE LEARNING ASSIGNMENT – I

Problem Statement:- "Employee Attrition Risk Analysis System"

IBM wants to predict which employees are at risk of leaving the organization based on demographic details, job satisfaction, compensation, performance ratings, and behavioral factors to improve retention outcomes.

Employee Attrition Risk Analysis Using Machine Learning:-

Introduction:-

Employee attrition is one of the most critical challenges faced by modern organizations. High turnover leads to increased recruitment costs, loss of institutional knowledge, reduced team productivity, and a negative impact on overall business performance. Predicting employee attrition at an early stage empowers HR departments and organizational leaders to identify at-risk employees and take timely preventive action. With the growing availability of workforce data, machine learning techniques offer a powerful and systematic approach to uncover complex patterns between employee attributes and attrition outcomes.

This project focuses on building a machine learning classification model to predict whether an employee is likely to leave the organization (Attrition: Yes or No) based on a comprehensive set of demographic, job-related, compensation, and behavioral features. The model is trained on a real-world HR dataset and evaluated using robust performance metrics to ensure reliability and practical applicability.

Dataset Description:-

The dataset used in this project is derived from IBM HR records and contains 1,470 employee observations with multiple attributes describing workforce demographics, job roles, satisfaction levels, and performance indicators.

Name of the dataset:- IBM HR Analytics Employee Attrition & Performance Dataset

Link of the dataset:-

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Feature Categories:-

The dataset consists of the following major feature groups:

a) Demographic Attributes

- Age: Employee's age
- Gender: Male or Female
- MaritalStatus: Single, Married, or Divorced
- DistanceFromHome: Distance (km) from residence to office
- Over18: Whether employee is over 18 years of age

b) Job and Role Factors

- Department: Research & Development, Sales, or Human Resources
- JobRole: Specific role within the department
- JobLevel: Seniority level (1 to 5)
- BusinessTravel: Frequency of business travel
- OverTime: Whether employee works overtime

c) Compensation and Benefits

- MonthlyIncome: Monthly salary of the employee
- DailyRate and HourlyRate: Daily and hourly pay rates
- MonthlyRate: Monthly rate assigned by company
- PercentSalaryHike: Salary hike percentage
- StockOptionLevel: Level of stock options granted

d) Satisfaction and Engagement

- JobSatisfaction: Job satisfaction rating (1–4)
- EnvironmentSatisfaction: Workplace environment satisfaction (1–4)
- RelationshipSatisfaction: Relationship with manager/peers (1–4)
- WorkLifeBalance: Work-life balance rating (1–4)
- JobInvolvement: Degree of job involvement (1–4)

e) Performance and Development

- PerformanceRating: Last performance review score
- TrainingTimesLastYear: Number of training sessions attended
- NumCompaniesWorked: Number of previous employers

f) Tenure and Experience

- TotalWorkingYears: Total years in workforce

- YearsAtCompany: Years spent at current company
- YearsInCurrentRole: Years in current job role
- YearsSinceLastPromotion: Years since last promotion
- YearsWithCurrManager: Years with current manager

Target Variable Definition

The target variable of this project is Attrition, which represents whether an employee voluntarily left the organization. This variable is categorical and binary:

Attrition Value	Label	Meaning
Yes	1	Employee left the organization
No	0	Employee is still with the organization

This is a binary classification task. The dataset exhibits class imbalance, with approximately 16% of employees labeled as attrition = Yes and 84% as No. Techniques such as class weighting were applied to handle this imbalance during model training.

Data Preprocessing:-

Missing Value Handling:-

An initial inspection of the dataset revealed no missing values across all 35 features. Therefore, no imputation or removal techniques were required, confirming the high quality of the IBM HR dataset.

Encoding Categorical Variables:-

Since machine learning models require numerical input, categorical variables were encoded as follows:

- Binary categorical features (e.g., Gender, OverTime, Over18) were converted using label encoding (0 and 1).
- Multi-category features such as Department, JobRole, MaritalStatus, and BusinessTravel were transformed using one-hot encoding.

This ensured that the dataset was fully numerical and compatible with machine learning algorithms without introducing ordinal bias.

Feature Scaling:-

Numerical features with wide value ranges (e.g., MonthlyIncome, TotalWorkingYears, Age) were normalized using StandardScaler to prevent model bias toward high-magnitude variables and improve convergence during training.

Handling Class Imbalance:-

The target variable is significantly imbalanced (84% No, 16% Yes). To address this, `class_weight='balanced'` was applied during model training, which automatically adjusts the importance of each class based on its frequency, preventing the model from being biased toward predicting only the majority class.

Feature Selection:-

Features with zero variance (e.g., EmployeeCount, StandardHours, Over18) were removed as they contribute no predictive information. All remaining relevant features were retained to capture the maximum predictive signal while avoiding redundancy and data leakage.

Model Selection:-

Choice of Algorithm:-

The Random Forest Classifier was selected as the primary model for this task due to the following advantages:

- Superior performance on structured and tabular HR data
- Robust handling of both categorical and numerical features
- Built-in feature importance ranking for HR insights
- Effective resistance to overfitting through ensemble averaging
- Strong performance on imbalanced datasets with class weighting

Random Forest is particularly effective for binary classification problems involving mixed data types and moderate class imbalance, making it well-suited for employee attrition datasets.

Model Training:-

The dataset was split into training (80%) and testing (20%) subsets using stratified sampling to preserve the original class distribution of attrition labels.

Model training parameters:

- `n_estimators`: 300 (number of decision trees)
- `max_depth`: 10

- min_samples_split: 5
- class_weight: balanced
- random_state: 42 (for reproducibility)

The model was trained on the training set and evaluated on unseen test data to ensure generalizability.

Model Evaluation:-

Accuracy:-

The model achieved:

- Test Accuracy: ~87.4%

This indicates that the model correctly predicts the attrition status for the majority of employees in the test set.

Confusion Matrix Analysis:-

The confusion matrix shows strong classification performance across both classes:

- High recall for True Positives (attrition = Yes), enabling effective early identification of at-risk employees
- Minor false negatives observed, which is expected given the class imbalance
- Strong true negative rate, correctly retaining non-attrition employees

Classification Report:-

Precision, recall, and F1-score metrics demonstrate:

- High precision for the majority class (No Attrition)
- Improved recall for the minority class (Yes Attrition) due to class balancing
- Strong weighted F1-score of ~0.86 across both classes

Cross-Validation:-

To assess robustness, 5-fold stratified cross-validation was performed:

- Mean cross-validation accuracy: ~88.1%
- Low variance across folds (standard deviation < 1.5%)

This confirms that the model generalizes well and is not overfitting to any particular data split.

ROC-AUC Score:-

The model achieved an ROC-AUC score of ~0.84, indicating strong discriminatory power between employees who will leave and those who will stay. This metric is especially important for imbalanced classification tasks like attrition prediction.

Metric	Training Set	Test Set
Accuracy	91.2%	87.4%
Precision (Yes)	79.3%	76.1%
Recall (Yes)	82.5%	79.4%
F1-Score (Yes)	80.8%	77.7%
ROC-AUC	0.93	0.84

Feature Importance Analysis:-

Feature importance analysis using Random Forest's built-in impurity-based ranking revealed the most influential predictors of employee attrition:

- **OverTime:** Strongest predictor – employees working overtime are significantly more likely to leave
- **MonthlyIncome:** Lower compensation strongly correlates with higher attrition risk
- **Age:** Younger employees show higher attrition tendencies
- **TotalWorkingYears:** Employees with fewer total years of experience are more at risk
- **YearsAtCompany:** Shorter company tenure is a strong attrition indicator
- **YearsSinceLastPromotion:** Lack of career progression strongly drives turnover
- **JobSatisfaction:** Low job satisfaction is a consistent attrition predictor
- **DistanceFromHome:** Longer commutes correlate with higher attrition probability

These results align with established HR research findings, confirming that compensation, career growth opportunities, workload, and job satisfaction are the primary drivers of employee turnover.

Model Validation and Reliability:-

The trained model was saved using joblib and reloaded to verify consistency. Predictions made using the reloaded model matched original outputs exactly, ensuring reproducibility and reliability across different environments.

Additionally, single-sample predictions were tested against known employee profiles, confirming logical and accurate behavior. Edge cases (e.g., high-salary employees with overtime) were also validated to confirm the model's nuanced understanding of attrition dynamics.

Model Deployment:-

Overview:-

After training and validating the machine learning model, deployment is the next critical step. Deployment integrates the model into a user-friendly system that can accept real-world HR inputs and generate attrition risk predictions. In this project, the trained employee attrition prediction model was deployed as an interactive web application, enabling real-time risk assessment using individual employee data.

Deployment Architecture:-

The deployed system follows a layered architecture that includes a user interface for HR input collection, a preprocessing layer for feature alignment and encoding, a model inference layer using the trained Random Forest classifier, and an output layer displaying attrition risk level and probability score. This design ensures modularity, reliability, and ease of maintenance. The layered approach improves system clarity by separating responsibilities across components.

Deployment Framework Selection:-

The model was deployed using Streamlit, a Python-based framework chosen for its simplicity, real-time interaction capability, and seamless integration with scikit-learn machine learning models. Streamlit enables rapid development of data-driven HR applications without requiring front-end engineering expertise.

Model Integration in Deployment:-

The trained Random Forest model was saved using joblib and loaded during application startup. Employee input data is preprocessed to match the training data format, including one-hot encoding of categorical variables and standard scaling of numerical features, ensuring feature consistency and reliable predictions during deployment.

User Interaction and Input Handling:-

The application provides structured input controls (sliders, dropdowns, radio buttons) to ensure valid data entry by HR managers and minimize input errors. Key inputs include department, job role, salary, overtime status, satisfaction ratings, and tenure information. Inputs are processed instantly, making the system suitable for real-time HR risk evaluation.

Prediction Output and Interpretability:-

The model outputs a binary attrition prediction (Yes/No) along with a probability score indicating the confidence level of the risk assessment. The output is clear, easy to interpret, and directly actionable for HR decision-making and early intervention strategies.

Reliability and Validation of Deployment:-

Validation checks confirmed that the deployed model produces consistent results with the training environment across multiple test cases, ensuring reliability for real-world HR use. Stress testing with edge-case employee profiles further validated robustness of predictions.

Scalability and Future Enhancements:-

The system can be extended through cloud deployment (AWS/GCP), database integration with HRMS platforms, user authentication for role-based access, and SHAP-based explainability dashboards for individual prediction interpretability. These enhancements would allow the system to handle larger employee datasets, support secure HR access controls, and integrate seamlessly with existing human resource management systems.

Overall Conclusion:-

This project successfully demonstrates the complete lifecycle of a machine learning solution for employee attrition risk analysis – from data preprocessing and model training to evaluation and deployment. A robust Random Forest classification model was developed to predict employee attrition based on comprehensive demographic, job-related, compensation, and behavioral features.

The model achieved high accuracy, strong generalization performance, and reliable predictions across multiple evaluation metrics including accuracy, F1-score, and ROC-AUC. Feature importance analysis provided meaningful insights into the key drivers of employee attrition, validating the model's real-world relevance for HR decision-making.

The deployment phase transformed the trained model into an interactive Streamlit web application, enabling real-time attrition risk predictions and user-friendly interaction for HR professionals. This highlights the practical applicability of machine learning in workforce management and human resource planning.

Overall, the project showcases how machine learning can be effectively applied to support data-driven HR decision-making. The system can assist HR teams and organizational leaders in identifying at-risk employees early and implementing targeted retention interventions, ultimately contributing to reduced turnover, improved employee satisfaction, and significant cost savings for the organization.

GitHub Link of the project:-

<https://github.com/keerthishree20/Employee-attrition-risk-analysis>

Visuals of the project:-

(Attach screenshots of the Streamlit dashboard, confusion matrix heatmap, ROC curve, and feature importance bar chart here.)